Polynomial Kernel Proof

$$[a^T b + 1]^2 = a^T a\, b^T b + 2a^T b + 1 = \varphi(a)^T \varphi(b)$$

Where: $\varphi(x) = [x_1^2, \sqrt{2}\, x_1 x_2, x_2^2, \sqrt{2}\, x_1, \sqrt{2}\, x_2, 1]^T$

$\varphi(x) \in \mathbb{R}^6$ — 2D input mapped to 6D feature space

$$\varphi(a)^T \varphi(b) = a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2 + 2a_1 b_1 + 2a_2 b_2 + 1$$

Numeric Example: $a = [1, 2]^T$, $b = [3, 4]^T$

$a^T b = 1 \cdot 3 + 2 \cdot 4 = 3 + 8 = 11$

$[a^T b + 1]^2 = (11 + 1)^2 = 12^2 = 144$

$\varphi(a) = [1, \sqrt{2} \cdot 2, 4, \sqrt{2} \cdot 1, \sqrt{2} \cdot 2, 1]^T = [1, 2\sqrt{2}, 4, \sqrt{2}, 2\sqrt{2}, 1]^T$

$\varphi(b) = [9, \sqrt{2} \cdot 12, 16, \sqrt{2} \cdot 3, \sqrt{2} \cdot 4, 1]^T = [9, 12\sqrt{2}, 16, 3\sqrt{2}, 4\sqrt{2}, 1]^T$

$\varphi(a)^T \varphi(b) = 1 \cdot 9 + (2\sqrt{2})(12\sqrt{2}) + 4 \cdot 16 + \sqrt{2} \cdot 3\sqrt{2} + (2\sqrt{2})(4\sqrt{2}) + 1 \cdot 1 = 9 + 48 + 64 + 6 + 16 + 1 = 144$

✓ $[a^T b + 1]^2 = 144 = \varphi(a)^T \varphi(b)$ — Kernel trick verified